



EXAM PAPERS PRACTICE

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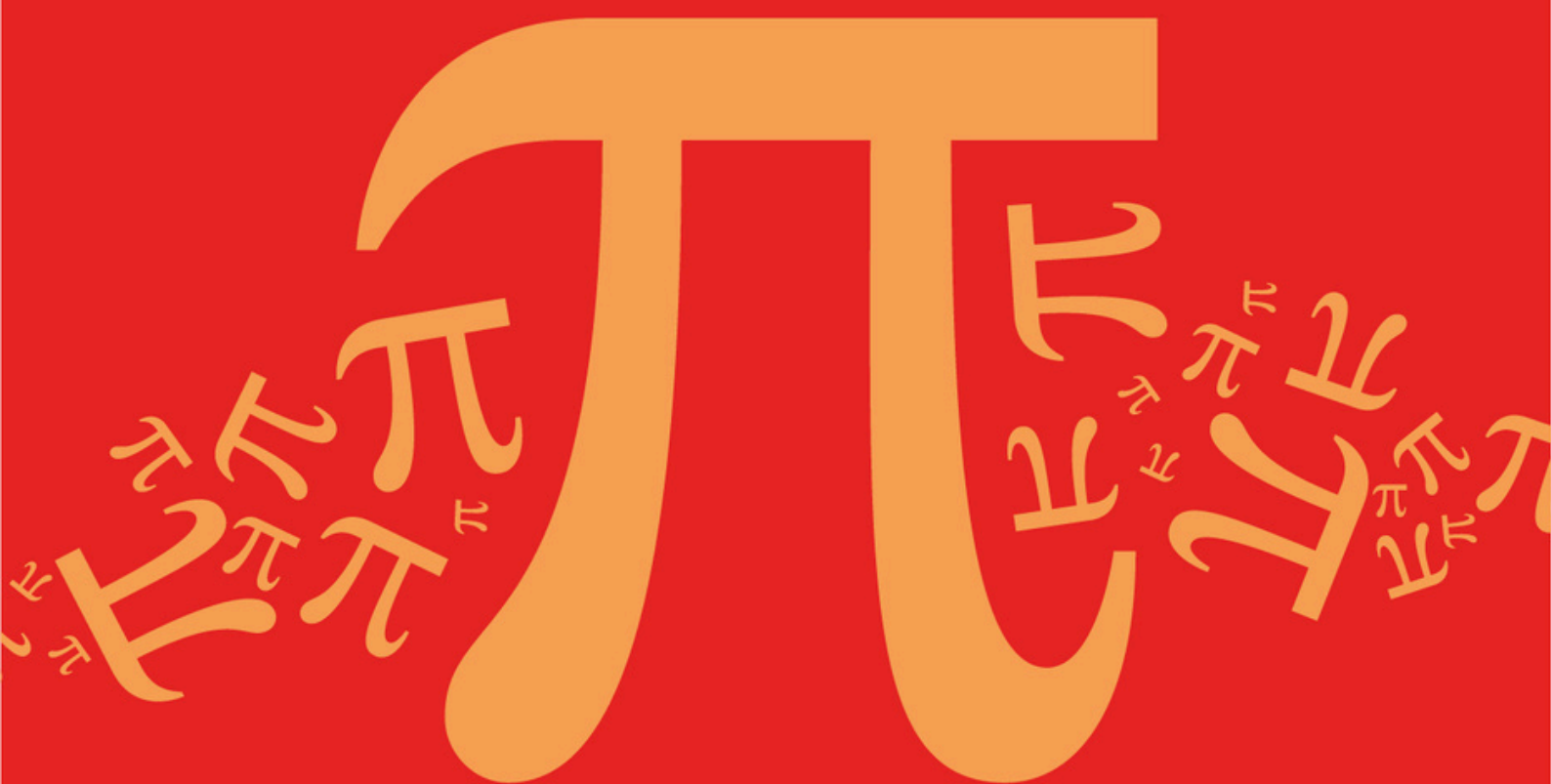
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C6: Singular Matrices (Properties of Singular Matrices)

Topic Questions

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Question Paper

Topic

C: Matrices

Sub topic

C6: Singular Matrices (Properties of Singular Matrices)

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

- (a) Show that the matrix $\begin{bmatrix} 5-k & 2 \\ k^3+1 & k \end{bmatrix}$ is singular when $k = 1$. (1)

- (b) Find the values of k for which the matrix $\begin{bmatrix} 5-k & 2 \\ k^3+1 & k \end{bmatrix}$ has a negative determinant.
Fully justify your answer. (5)
- (Total 6 marks)

Q2.

- (a) By direct expansion, or otherwise, show that the value of $\begin{vmatrix} -2 & 1 & 2k \\ -1 & 1 & k+1 \\ 2 & k-1 & 1 \end{vmatrix}$ is independent of k . (4)

- (b) (i) State, with a reason, whether the equations

$$-2x + y + 6z = 1$$

$$-x + y + 4z = 0$$

$$2x + 2y + z = -1$$

are consistent or inconsistent. (2)

- (ii) The three equations given in part (b)(i) are the Cartesian equations of three planes.

State the geometrical configuration of these three planes. (1)

(Total 7 marks)

Q3.

The system of equations S is given in terms of the real parameters a and b by

$$2x + y + 3z = a + 1$$

$$5x - 2y + (a + 1)z = 3$$

$$ax + 2y + 4z = b$$

- (a) Find the two values of a for which S does not have a unique solution. (4)

- (b) In the case when $a = 2$, determine the value of b for which S has infinitely many solutions.

(4)

(Total 8 marks)

Q4.

- (a) Determine the two values of k for which the system of equations

$$\begin{aligned} x - 2y + kz &= 5 \\ (k+1)x + 3y &= k \\ 2x + y + (k-1)z &= 3 \end{aligned}$$

does not have a unique solution.

(4)

- (b) Show that this system of equations is consistent for one of these values of k , but is inconsistent for the other.

(You are not required to find any solutions to this system of equations.)

(8)

(Total 12 marks)

Q5.

Let $\mathbf{P} = \begin{bmatrix} 1 & 4 & 2 \\ -1 & 2 & 6 \end{bmatrix}$ and $\mathbf{Q} = \begin{bmatrix} k & 1 \\ 2 & -1 \\ 3 & 1 \end{bmatrix}$, where k is a constant.

- (a) Determine the product matrix $\mathbf{PQ}$, giving its elements in terms of k where appropriate.

(3)

- (b) Find the value of k for which $\mathbf{PQ}$ is singular.

(2)

(Total 5 marks)

Q6.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are given by

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 2 & t \end{bmatrix}$$

- (a) Find, in terms of t , the matrices:

(i) $\mathbf{AB}$;

(3)

(ii) $\mathbf{BA}$.

- (2)
- (b) Explain why $\mathbf{AB}$ is singular for all values of t . (1)
- (c) In the case when $t = -2$, show that the transformation with matrix $\mathbf{BA}$ is the combination of an enlargement, E, and a second transformation, F. Find the scale factor of E and give a full geometrical description of F. (6)
- (Total 12 marks)

Q7.

The matrices $\mathbf{P}$ and $\mathbf{Q}$ are given by

$$\mathbf{P} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & t & -2 \\ 3 & 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ 11 & -1 & -7 \end{bmatrix}$$

where t is a real constant.

- (a) Find the value of t for which $\mathbf{P}$ is singular. (2)
- (b) (i) Determine the matrix $\mathbf{R} = \mathbf{PQ}$, giving its elements in terms of t where appropriate. (3)
- (ii) Find the value of t for which $\mathbf{R} = k\mathbf{1}$, for some integer k . (2)
- (iii) Hence find the matrix $\mathbf{Q}^{-1}$. (1)
- (c) In the case when $t = -3$, describe the geometrical transformation with matrix $\mathbf{R}$. (2)
- (Total 10 marks)